

MALYE KARMAKULY, Russian Federation

WMO: 207440

Lat: 72.373N

Lon: 52.717E

Elev: 19

StdP: 101.10

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-27.1	-25.5	-31.2	0.2	-26.7	-29.4	0.3	-25.1	33.3	-22.7	30.7	-23.7	8.7	100	1.107

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.9	16.4	12.7	14.0	11.1	12.0	9.8	13.2	15.7	11.4	13.6	9.9	11.7	8.1	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.7	8.6	14.9	10.0	7.6	12.8	8.6	7.0	11.0	37.3	15.7	32.8	13.9	29.2	11.8	18.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.3	22.3	19.0	DB	-27.3	18.4	2.5	3.6	-29.1	21.0	-30.6	23.1	-32.0	25.1	-33.7	27.6	(4)
(5)			WB	-27.3	14.4	3.3	2.1	-29.7	16.0	-31.6	17.2	-33.5	18.4	-36.0	20.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-3.6	-12.6	-13.2	-11.6	-8.3	-3.2	3.0	7.7	7.0	4.3	-1.2	-6.3	-9.3		
	DBStd	8.94	7.16	7.39	6.34	4.96	3.99	3.37	3.81	2.58	2.66	4.22	5.23	6.46		
	HDD10.0	4991	699	650	668	550	410	213	93	101	173	347	490	597		
	HDD18.3	8000	957	883	927	800	668	460	329	352	422	606	740	856		
	CDD10.0	33	0	0	0	0	0	3	23	7	1	0	0	0		
	CDD18.3	1	0	0	0	0	0	0	1	0	0	0	0	0		
	CDH23.3	1	0	0	0	0	0	0	1	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	7.7	9.3	9.7	8.8	8.2	6.9	6.8	5.8	6.2	6.5	7.5	8.3	9.0		
	PrecAvg	343	27	20	20	17	17	24	43	39	47	34	22	28		
(15) Precipitation	PrecMax	480	86	93	70	64	40	47	121	108	127	87	52	105		
	PrecMin	238	4	3	5	5	4	2	1	1	2	9	2	4		
	PrecStd	67	22	17	14	12	10	12	27	28	26	19	13	21		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.7	0.2	0.6	1.9	9.2	16.1	20.7	16.3	12.2	7.4	2.1	1.3		
		MCWB	0.2	-0.3	-0.2	0.8	5.7	11.6	15.4	12.8	9.9	6.3	1.3	0.6		
	2%	DB	-0.2	-0.7	-0.3	0.5	4.8	13.0	17.9	13.8	10.4	5.8	1.3	0.6		
		MCWB	-0.7	-1.1	-0.9	-0.2	2.7	9.4	13.7	11.1	8.8	4.9	0.6	0.1		
	5%	DB	-1.2	-1.6	-1.1	-0.3	2.7	10.2	15.8	12.2	9.1	4.8	0.7	0.0		
		MCWB	-1.8	-2.1	-1.8	-0.9	1.4	7.0	12.6	10.3	8.0	4.2	0.0	-0.5		
	10%	DB	-2.6	-3.0	-2.2	-1.2	1.6	7.8	13.5	10.8	8.0	3.8	0.0	-0.8		
		MCWB	-3.2	-3.5	-2.8	-1.8	0.6	5.6	10.9	9.0	7.1	3.1	-0.6	-1.3		
	0.4%	WB	0.4	-0.2	0.1	1.1	5.9	12.4	16.0	13.3	10.4	6.5	1.6	0.9		
		MCDB	0.5	-0.1	0.3	1.7	9.2	15.1	19.6	15.3	11.8	7.2	1.9	1.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-0.6	-1.0	-0.8	0.0	2.9	9.8	14.4	11.7	9.1	5.1	0.8	0.1		
		MCDB	-0.3	-0.7	-0.4	0.4	4.2	13.0	17.4	13.3	10.2	5.7	1.2	0.4		
	5%	WB	-1.7	-2.1	-1.7	-0.8	1.6	7.2	12.8	10.4	8.0	4.2	0.2	-0.5		
		MCDB	-1.2	-1.6	-1.2	-0.4	2.5	9.6	15.5	11.9	8.8	4.8	0.6	-0.1		
	10%	WB	-3.2	-3.5	-2.8	-1.8	0.7	5.7	11.0	9.2	7.1	3.2	-0.5	-1.3		
		MCDB	-2.6	-3.0	-2.2	-1.2	1.5	7.6	13.3	10.6	8.0	3.7	0.0	-0.8		
	5% DB	MDBR	4.9	4.9	4.9	5.0	3.7	3.1	3.9	3.3	2.9	3.0	4.0	4.7		
		MCDBR	4.5	4.8	4.5	4.5	4.4	6.5	7.1	5.7	3.9	2.6	3.0	4.0		
		MCWBR	4.7	4.9	4.5	4.4	3.2	4.5	4.5	4.2	3.1	2.6	3.1	4.1		
		MCWBR	4.5	4.8	4.4	4.5	4.1	6.0	7.1	5.2	3.5	2.5	3.0	4.0		
(35) Mean Daily Temperature Range	5% WB	MCWBR	4.7	5.0	4.6	4.6	3.1	4.4	4.7	4.1	3.0	2.6	3.2	4.1		
	taub		N/A	0.229	0.287	0.313	0.329	0.325	0.348	0.330	0.299	N/A	N/A	N/A		
		taud	N/A	1.928	2.076	2.136	2.207	2.361	2.410	2.452	2.427	N/A	N/A	N/A		
(42) Clear-Sky Solar Irradiance	Ebn at Noon		0	344	651	787	831	855	815	783	682	N/A	0	0		
		Edh at Noon	0	32	73	101	109	97	90	75	54	N/A	0	0		
	RadAvg		0.00	0.17	1.40	3.60	5.46	5.56	4.64	2.90	1.43	0.36	0.00	0.00		
(44) All-Sky Solar Radiation	RadStd		0.00	0.02	0.18	0.26	0.33	0.56	0.43	0.31	0.11	0.05	0.00	0.00		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.57	+2.60	+2.77	+1.75	+1.70	+1.82	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page